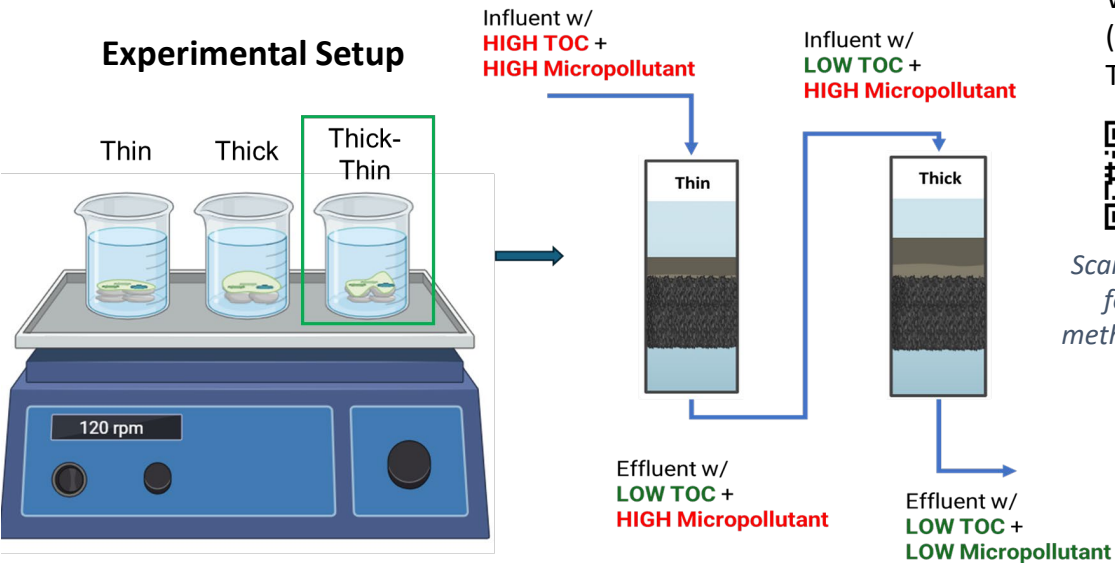


RO1: Redesign biofilters to support thick-thin biofilms, then investigate the impact of thickness on microbial diversity and TOC/MP removal.

Determine the influence of high and low TOC concentrations on biofilm thickness, biodiversity, and micropollutant degradation.

Experimental Setup**RO1 Batch studies**

Separate tests using biofilm Z-carriers and media are needed to distinguish adsorption from biodegradation.

Biofilm carrier

Z series

**RO2 Small-scale column studies**

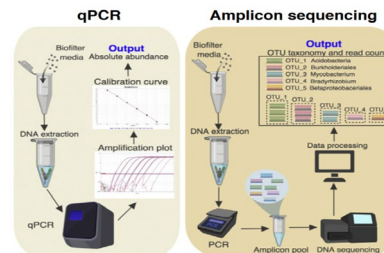
GAC media
(w/ adsorptive capacity)

Data Collection

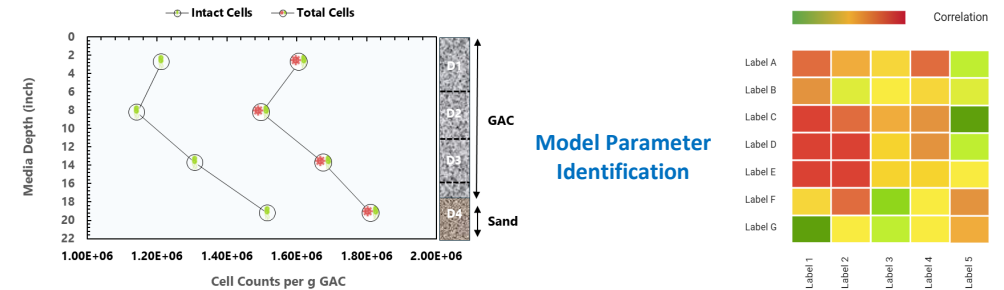
Water quality parameters
(pH, temperature, DO,
TOC, MP,...)



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methods paper

**Flow Cytometry
(cell counts)****Confocal microscopy
(thickness in mm)****qPCR and 16s rRNA gene
sequencing
(abundance and identity
of microorganisms)****RO2: Quantify the impact of key operational parameters and evaluate how their interactions collectively influence MP removal efficiency.**

	BR1 Thin (Single)		BR2 Thick (Single)		BR3 Thin-Thick (Combined)		BR1-BR2 Thin-Thick (2-Stage)	
	Low TOC	High TOC	Low TOC	High TOC	Low TOC	High TOC	Low TOC	High TOC
Biofilm Density (cells/mL)	###	###	###	###	###	###	###	###
Biofilm Thickness (μm)	Z200	Z200	Z400	Z400	Z200 (50%) Z400 (50%)	Z200 (50%) Z400 (50%)	Stage 1: Z200 Stage 2: Z400	Stage 1: Z200 Stage 2: Z400
TOC Removal (mg/L)	Good	Good	Good	Good	Good	Good	Good	Good
Micropollutant Removal (ng/L-μ/L)	Bad	Bad	Good	Bad	Good	Bad	Good	Good

**RO3: Develop a hybrid model, combining mechanistic and data-driven approaches, to overcome individual limitations.****Model Development**